

Setting Windows Console Size

These notes address my investigation as to why I cannot set the size of my windows console program.

Introduction

I am trying to complete my notes on Windows Console Programming. I am first completing the article at <https://www.benryves.com/tutorials/winconsole/>. The program is not working as expected because the window is opening with more than 80 columns even when I set the number of columns to 80 and the number of lines to 50.

Current Code

The code being tested:

```
#include <stdlib.h>
#include <stdio.h>
#include <Windows.h>
#include <Tchar.h>

HANDLE wHnd;    // Handle to write to the console.
HANDLE rHnd;    // Handle to read from the console.

void FreezeWindow() {
    // Disable the window's ability to be resized:
    HWND consoleWindow = GetConsoleWindow(); // Get handle to the console window
    // Remove the sizing border and maximize button
    LONG style = GetWindowLong(consoleWindow, GWL_STYLE);
    style &= ~(WS_SIZEBOX | WS_MAXIMIZEBOX);
    SetWindowLong(consoleWindow, GWL_STYLE, style);

    // Apply the changes
    SetWindowPos(consoleWindow, nullptr, 0, 0, 0, 0,
        SWP_NOMOVE | SWP_NOSIZE | SWP_NOZORDER | SWP_FRAMECHANGED);
}

void SetConsoleSize(HANDLE hConsole, int width, int height) {
    // Create a COORD to hold the buffer size:
    COORD bufferSize = { (SHORT)width, (SHORT)height };
    SetConsoleScreenBufferSize(hConsole, bufferSize);

    // Set up the required window size:
    SMALL_RECT windowSize = { 0, 0, (SHORT)(width - 1), (SHORT)(height - 1) };

    // Change the console window size:
    SetConsoleWindowInfo(hConsole, TRUE, &windowSize);
}
```

```

int _tmain(int argc, _TCHAR* argv[]) {
    // Set up the handles for reading/writing:
    wHnd = GetStdHandle(STD_OUTPUT_HANDLE);
    rHnd = GetStdHandle(STD_INPUT_HANDLE);

    // Set the window and buffer size
    SetConsoleSize(wHnd, 80, 50);
    system("mode con: cols=80 lines=50");

    // Change the window title:
    SetConsoleTitle(TEXT("Win32 Console Control Demo"));
    FreezeWindow();
    // Set up the character buffer:
    CHAR_INFO consoleBuffer[80 * 50];

    // We'll fill the console buffer with random data:
    for (int y = 0; y < 50; ++y) {
        for (int x = 0; x < 80; ++x) {

            // An ANSI character is in the range 0-255,
            // so use % to keep it in this range.
            consoleBuffer[x + 80 * y].Char.AsciiChar = rand() % 256;

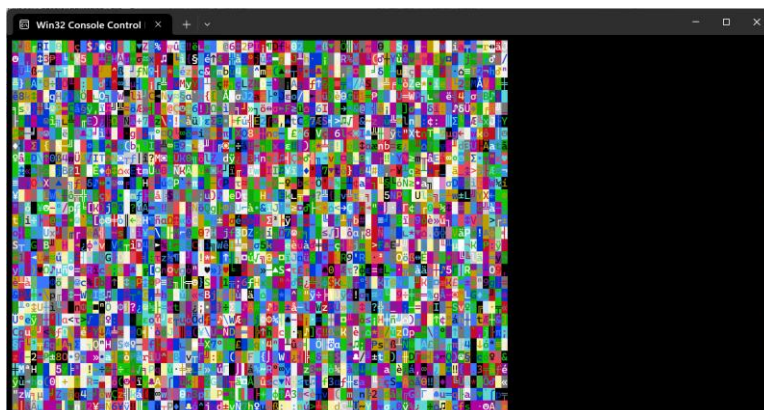
            // The colour is also in the range 0-255,
            // as it is a pair of 4-bit colours.
            consoleBuffer[x + 80 * y].Attributes = rand() % 256;
        }
    }

    // Set up the positions:
    COORD charBufSize = { 80, 50 };
    COORD characterPos = { 0, 0 };
    SMALL_RECT writeArea = { 0, 0, 79, 49 };

    // Write the characters:
    WriteConsoleOutputA(wHnd, consoleBuffer, charBufSize, characterPos, &writeArea);
    getchar(); // pauses the screen so we can see the result
    return 0;
}

```

The window that opened looked odd:



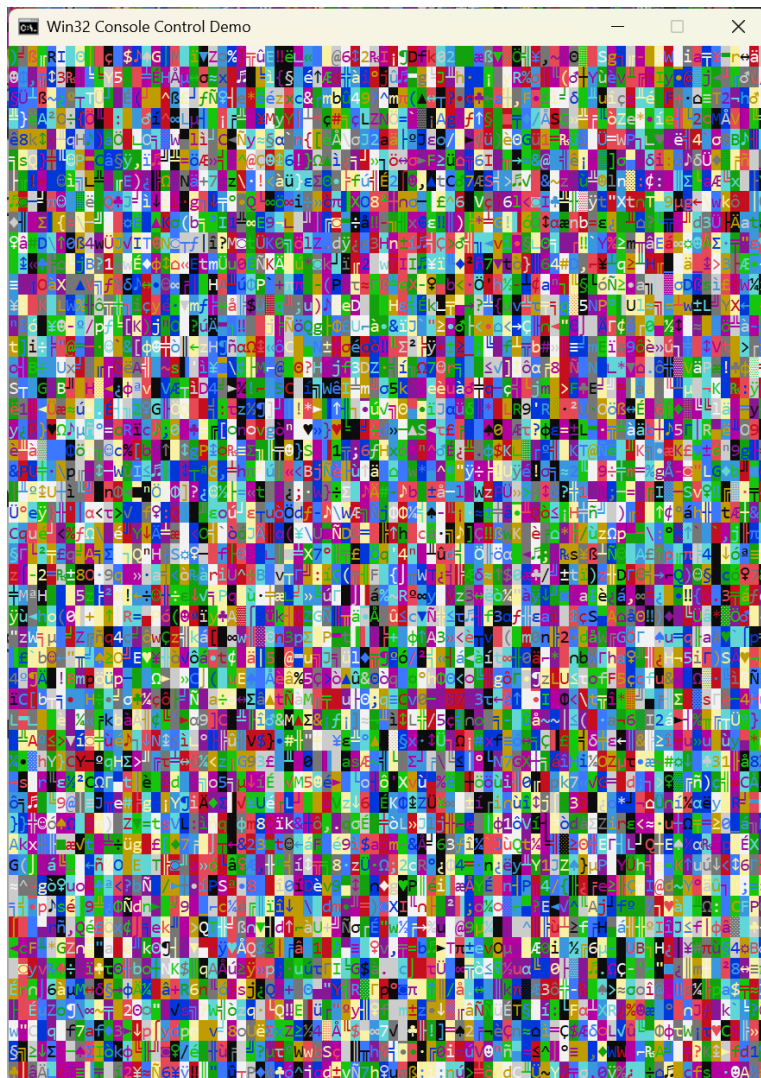
It opens with a lot of space on the right side. I added code to disable re-sizing of the window because then it really messes with the terminal window.

Solution #1

Run the application outside the IDE.

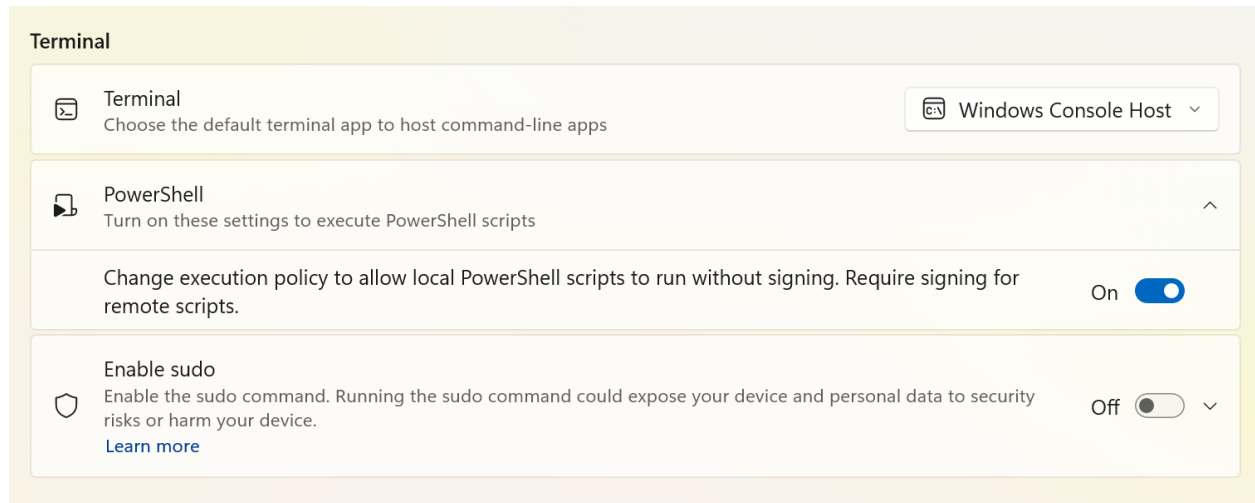
1. Navigate to the application.exe
2. Enter: `cohost.exe cmd.exe /c YourApplication.exe`

We now see the window as expected:



Solution #2

Open System → Terminal



Select “Windows Console Host” (it was originally set to “Let Windows Decide”).

Now you can run the program from the IDE or command line and see the program window correctly set as in Solution #1.

